

CSE 311:

Data Communication

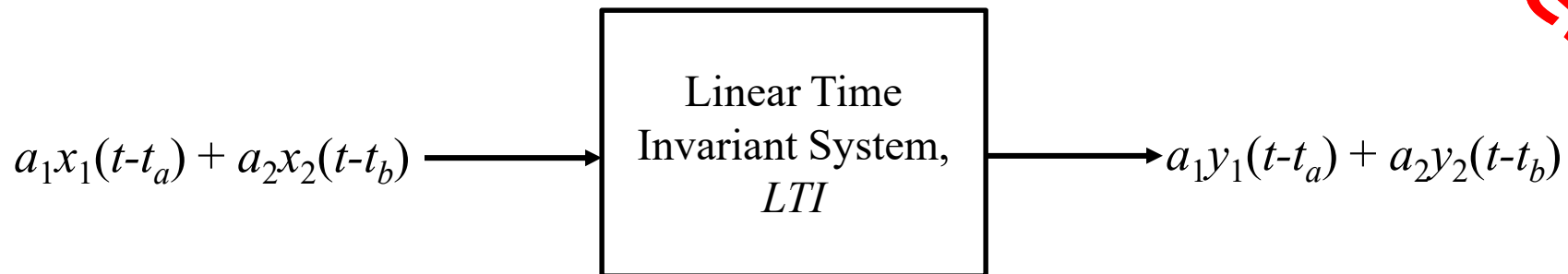
Instructor:
Dr. Md. Monirul Islam

Analysis of Signals and Systems

Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

Review

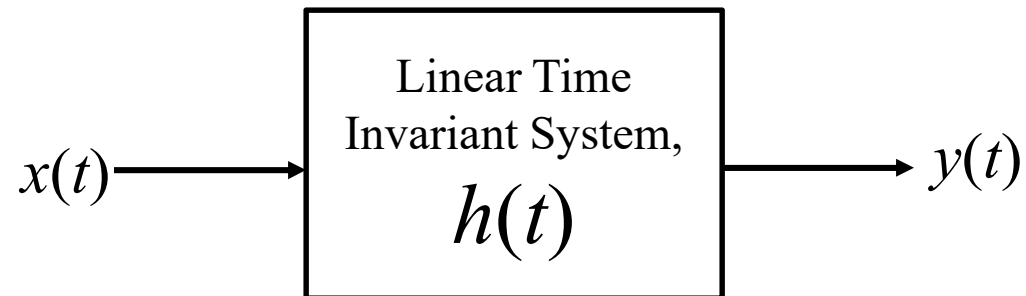


$$y(t) = LTI(x(t))$$

$$a_1y_1(t-t_a) + a_2y_2(t-t_b) = LTI(a_1x_1(t-t_a) + a_2x_2(t-t_b))$$

Signal Transmission through Linear Time Invariant System

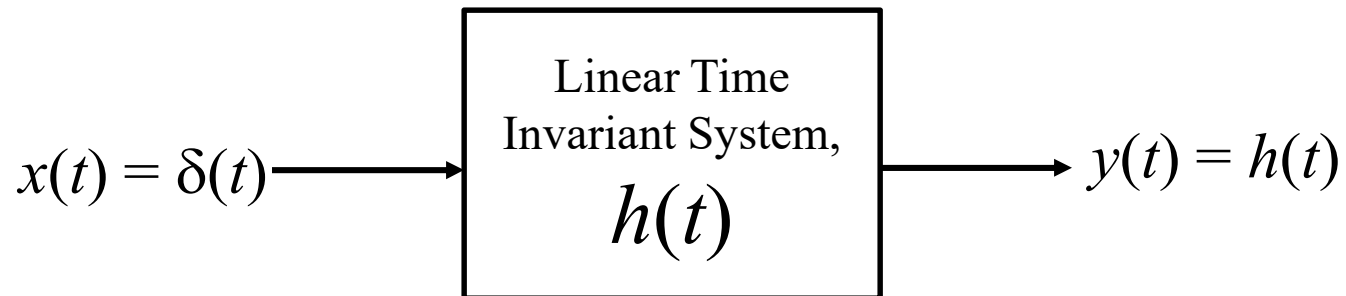
Linear Time Invariant System



LTI system is described by impulse response, $h(t)$

Signal Transmission through Linear Time Invariant System

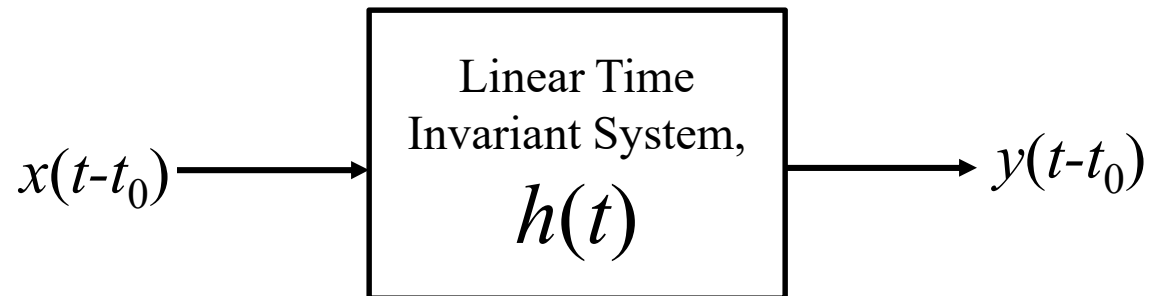
Linear Time Invariant System



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Signal Transmission through Linear Time Invariant System

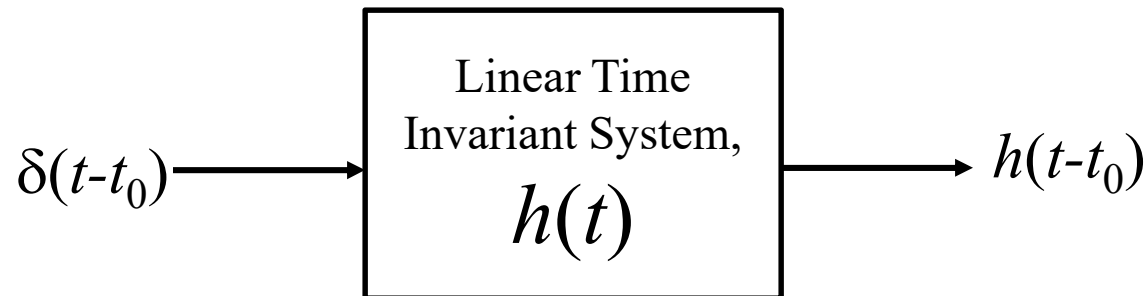
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Signal Transmission through Linear Time Invariant System

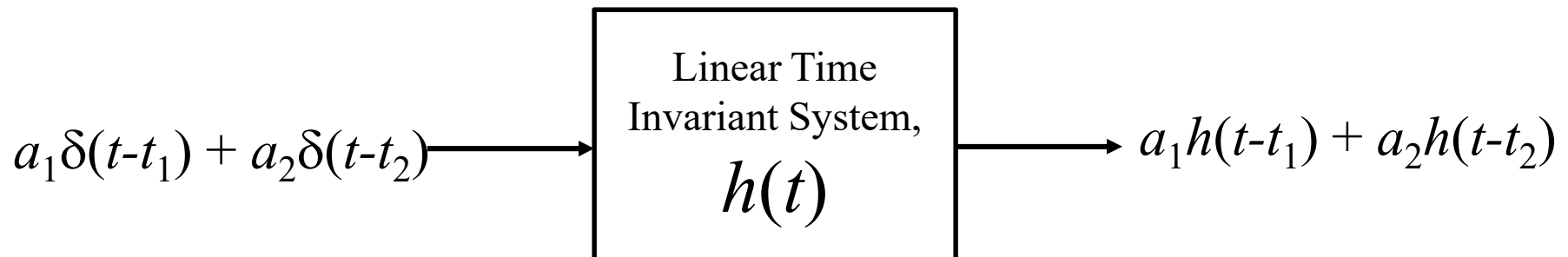
Linear Time Invariant System



LTI system is described by impulse response, $h(t)$

Signal Transmission through Linear Time Invariant System

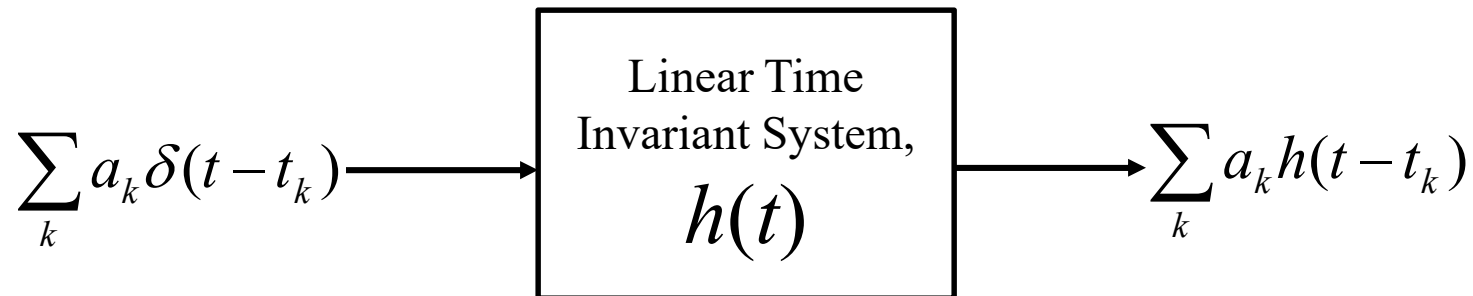
Linear Time Invariant System



LTI system is described by impulse response, $h(t)$

Signal Transmission through Linear Time Invariant System

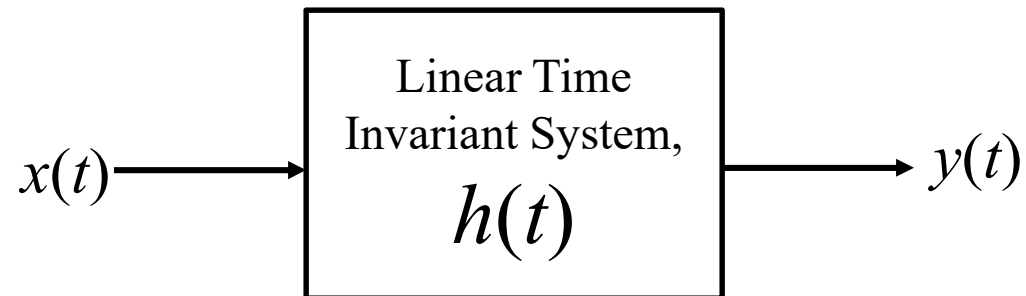
Linear Time Invariant System



LTI system is described by impulse response, $h(t)$

Signal Transmission through Linear Time Invariant System

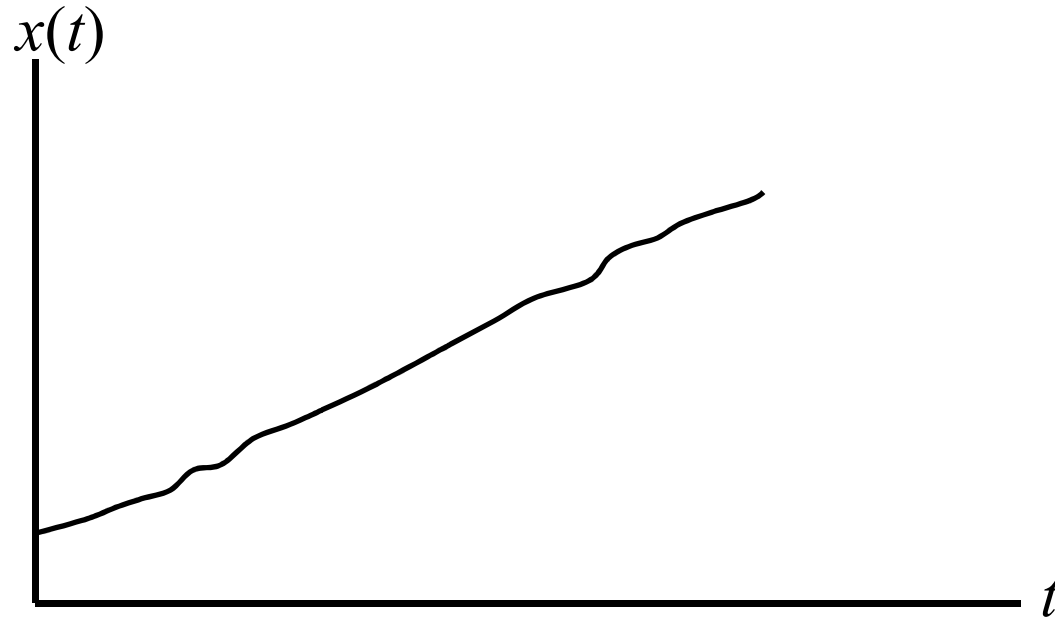
Linear Time Invariant System



We will find relation betn $x(t)$ and $y(t)$

Signal Transmission through Linear Time Invariant System

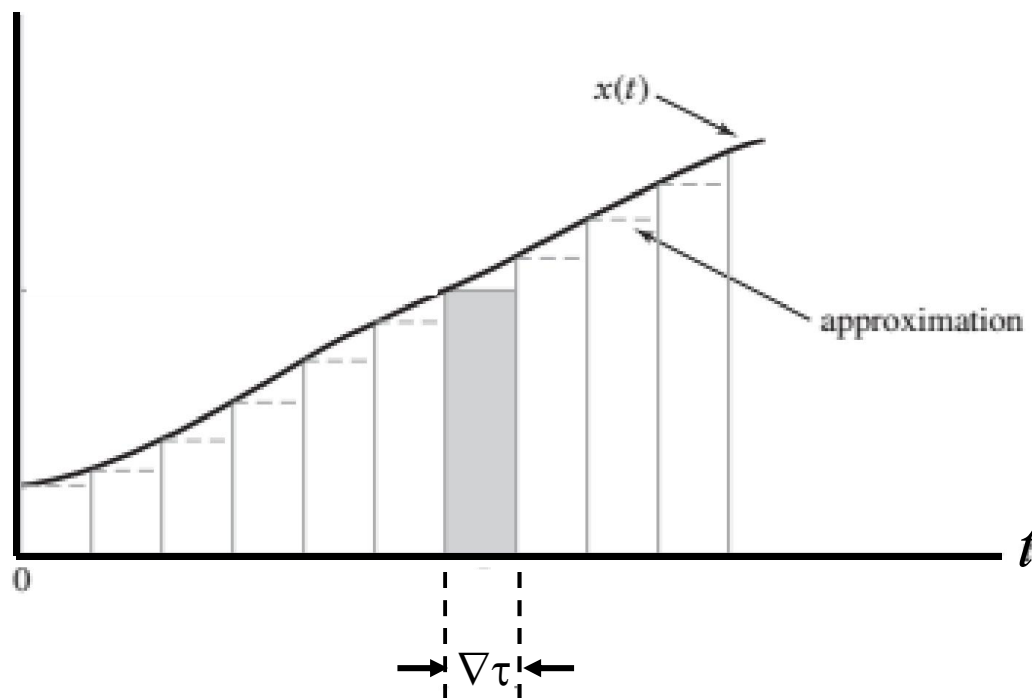
Linear Time Invariant System



Let the input signal be $x(t)$

Signal Transmission through Linear Time Invariant System

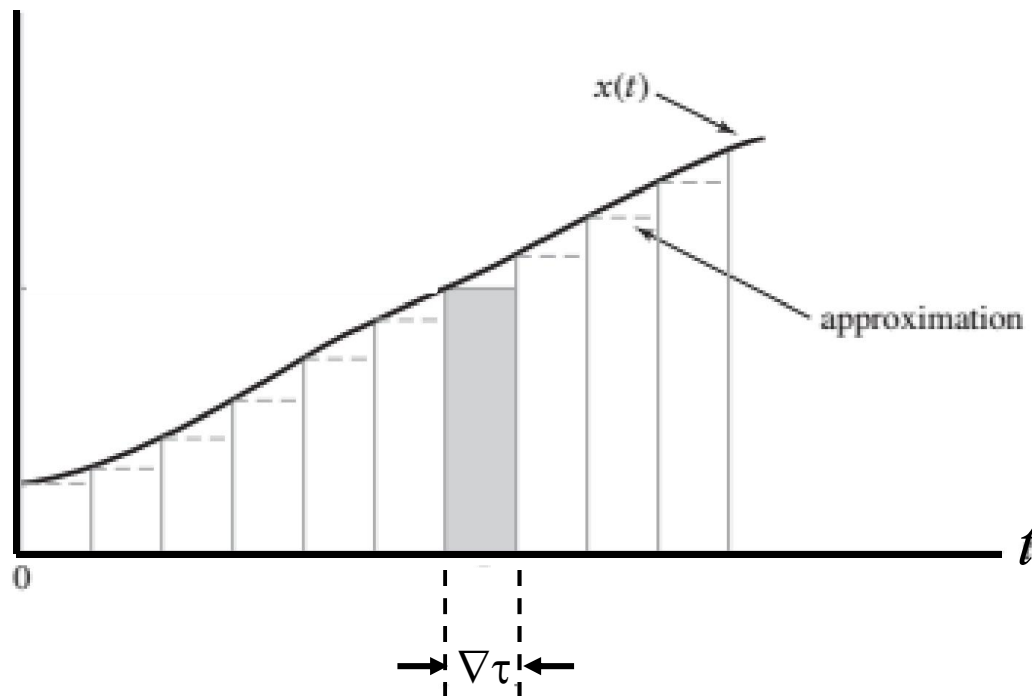
Linear Time Invariant System



Stair case approximation of $x(t)$ using narrow rectangular pulses

Signal Transmission through Linear Time Invariant System

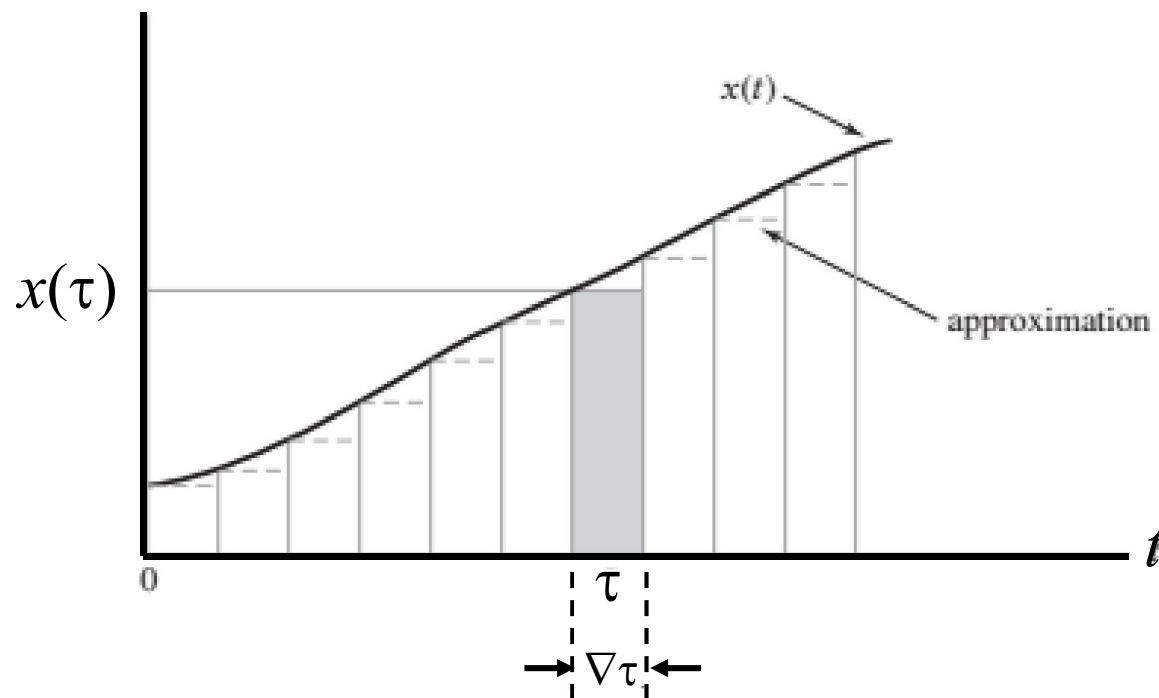
Linear Time Invariant System



As $\Delta\tau$ approaches to 0, pulse approaches to $\delta(\cdot) \times \text{height} \times \Delta\tau$

Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

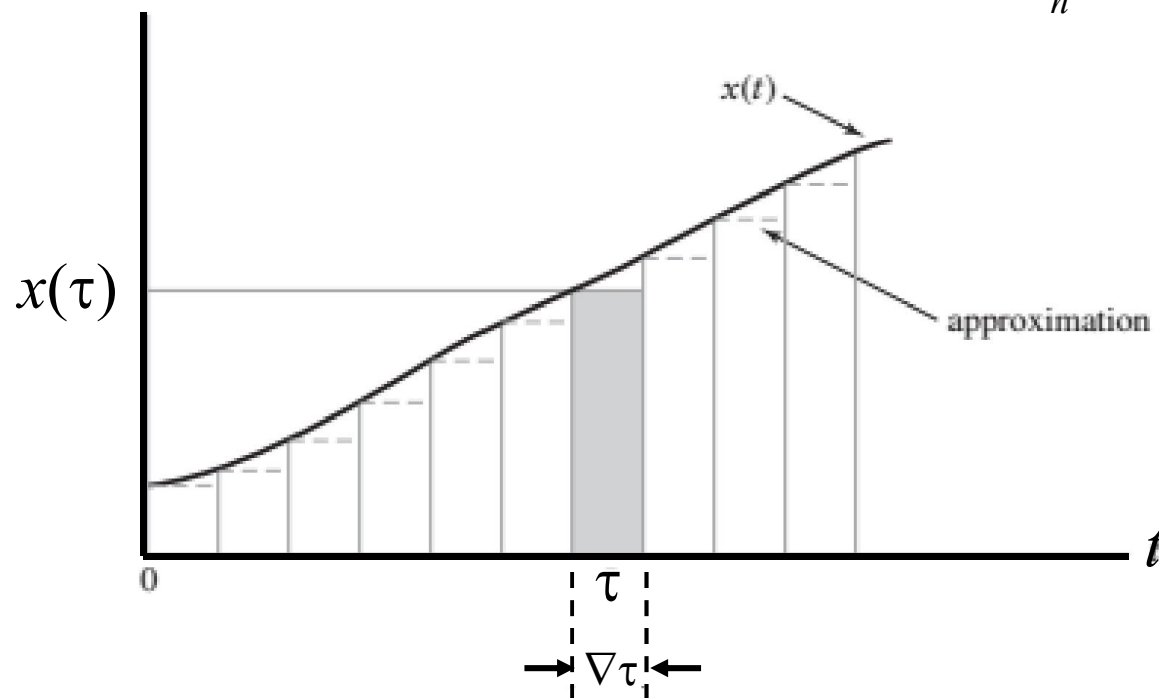


At $t = \tau$, pulse approaches to $\delta(t - \tau)x(\tau) \Delta\tau$

Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

$$x(t) = \sum_n \delta(t - n\nabla\tau) x(n\nabla\tau) \nabla\tau$$

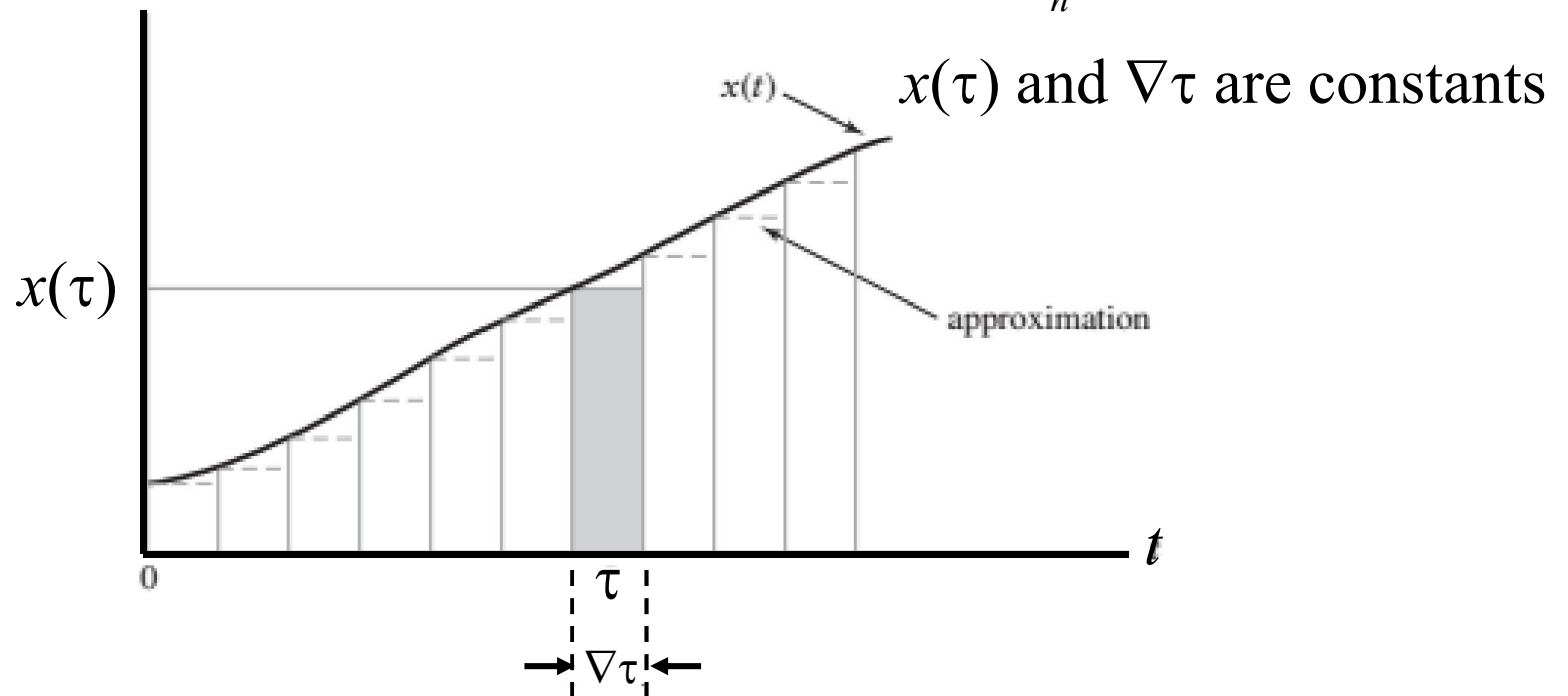


At $t = \tau$, pulse approaches to $\delta(t - \tau) x(\tau) \nabla\tau$

Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

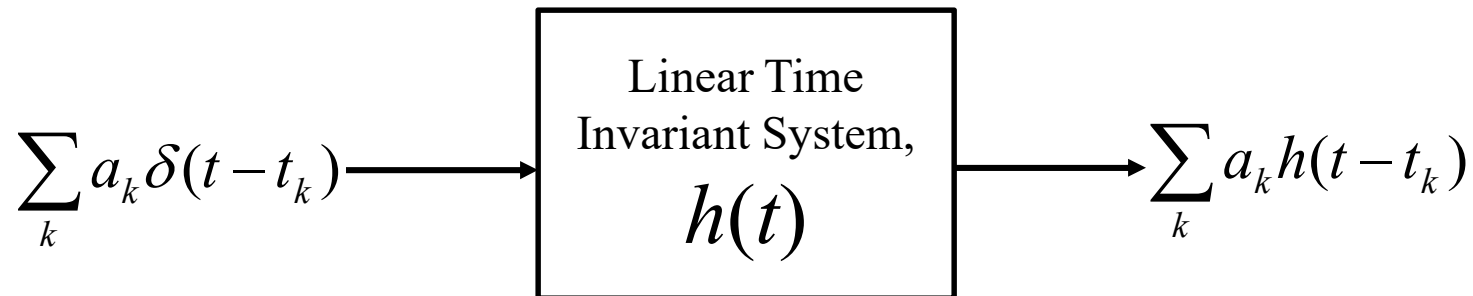
$$x(t) = \sum_n \delta(t - n\nabla\tau) x(n\nabla\tau) \nabla\tau$$



At $t = \tau$, pulse approaches to $\delta(t - \tau) x(\tau) \nabla\tau$

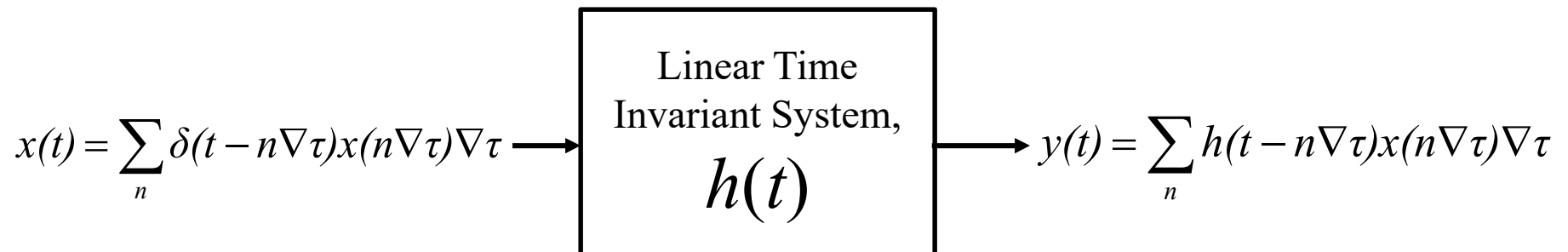
Signal Transmission through Linear Time Invariant System

Recall Linear Time Invariant System



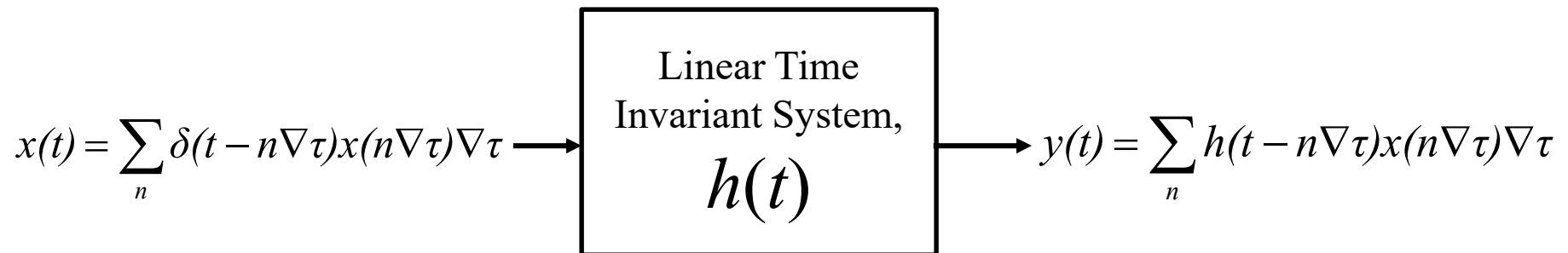
Signal Transmission through Linear Time Invariant System

Linear Time Invariant System



Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

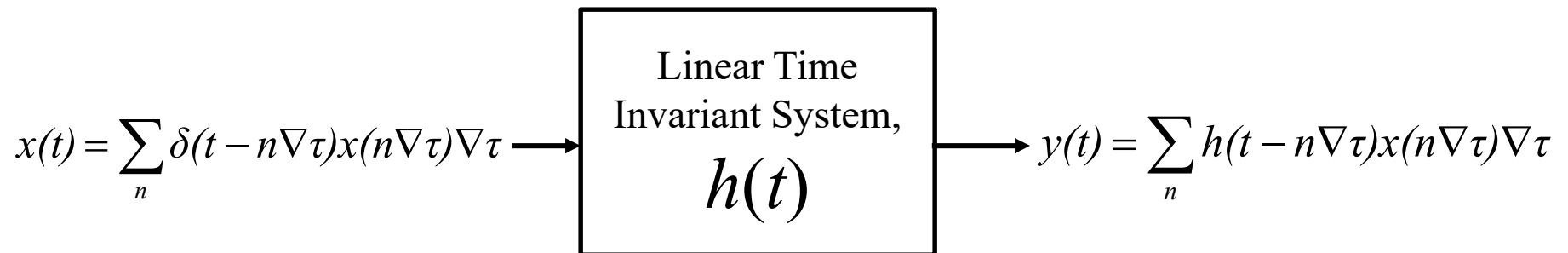


As $\nabla\tau$ approaches zero,

$$y(t) = \sum_n h(t - n\nabla\tau) x(n\nabla\tau) \nabla\tau = \sum_{\nabla\tau} h(t - \tau) x(\tau) \nabla\tau$$

Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

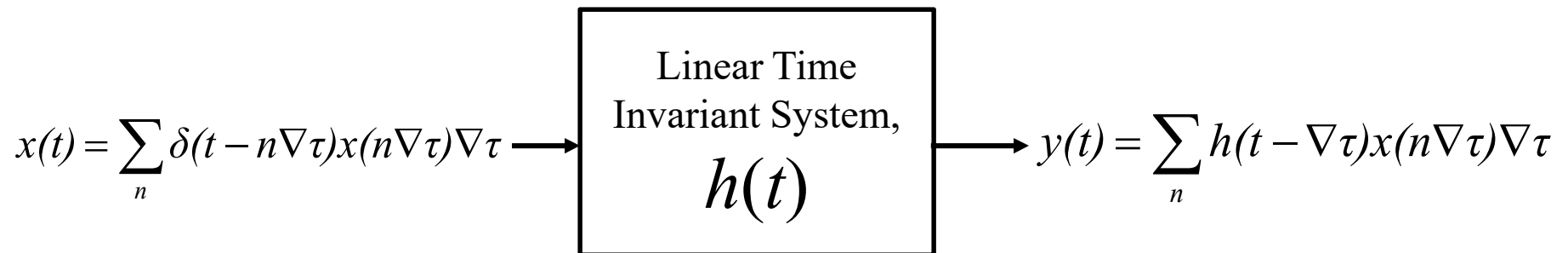


As $\nabla\tau$ approaches zero,

$$y(t) = \sum_n h(t - n\nabla\tau)x(n\nabla\tau)\nabla\tau = \sum_{\nabla\tau} h(t - \tau)x(\tau)\nabla\tau = \int_{-\infty}^{\infty} h(t - \tau)x(\tau)d\tau$$

Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

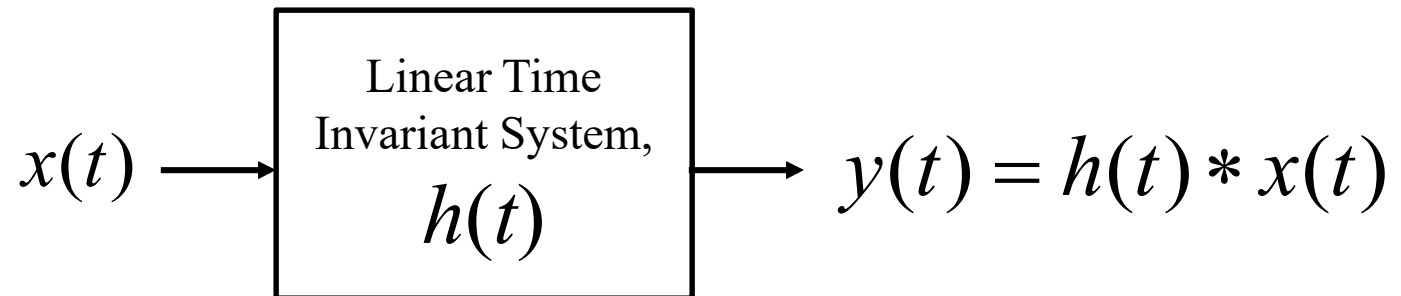


$y(t) = \int_{-\infty}^{\infty} h(t - \tau) x(\tau) d\tau$ is **convolution integral** and can be written as

$$y(t) = h(t) * x(t)$$

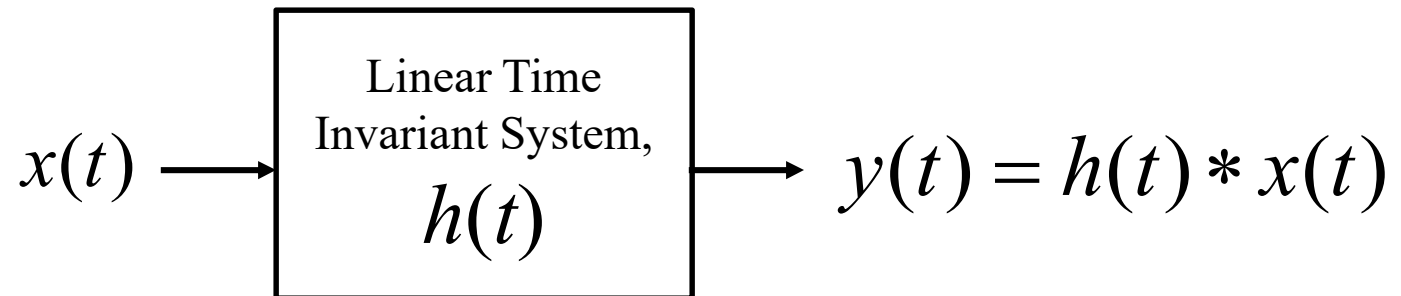
Signal Transmission through Linear Time Invariant System

Linear Time Invariant System



Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

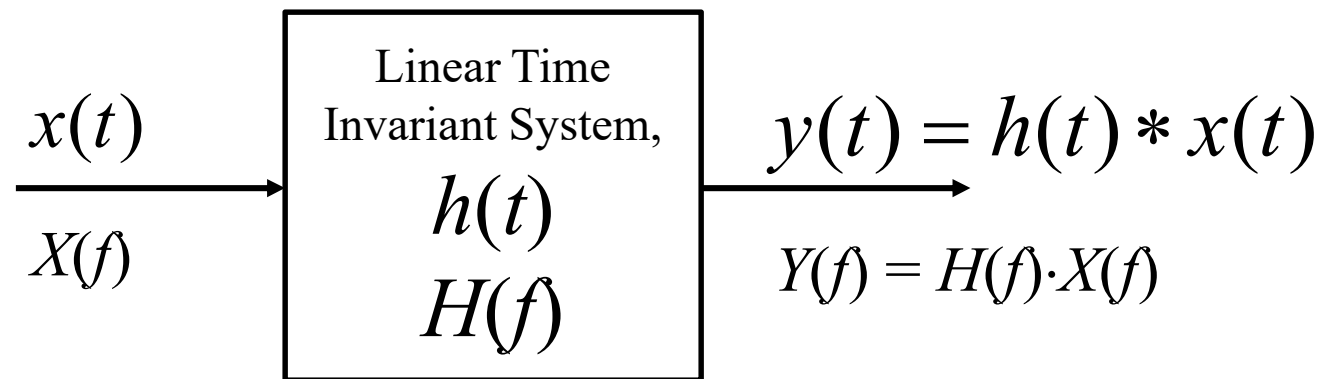


Now Recall

$$g_1(t) * g_2(t) \Leftrightarrow G_1(f)G_2(f)$$

Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

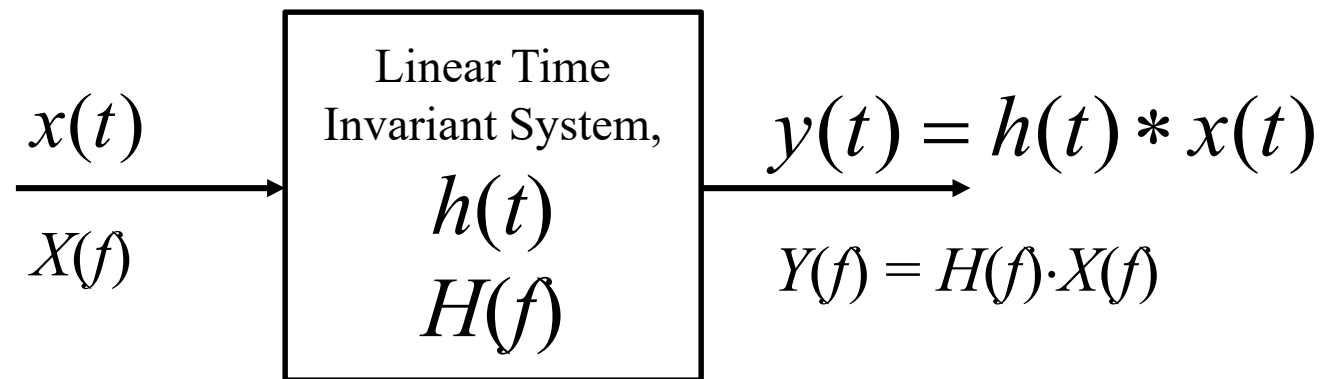


$H(f)$ is

- Fourier transform of $h(t)$
- the **frequency response** of LTI system
- a **transfer function**, because it transfers or passes desired frequencies of $X(f)$
- a **complex** function

Signal Transmission through Linear Time Invariant System

Linear Time Invariant System

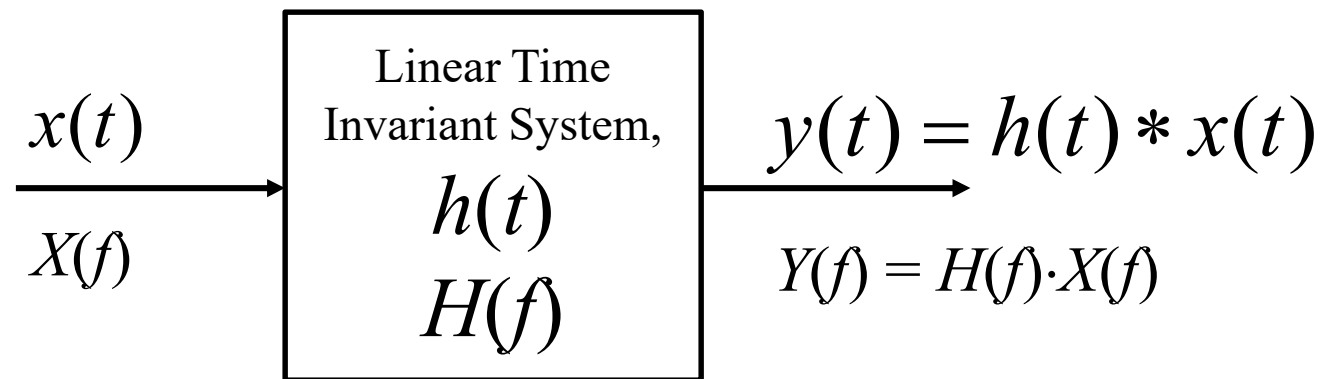


$$H(f) = |H(f)|e^{j\theta_h(f)}$$

- $|H(f)|$ is **amplitude response** of LTI system
- $\theta_h(f)$ is **phase response** of LTI system, measured in radians

Signal Transmission through Linear Time Invariant System

Linear Time Invariant System



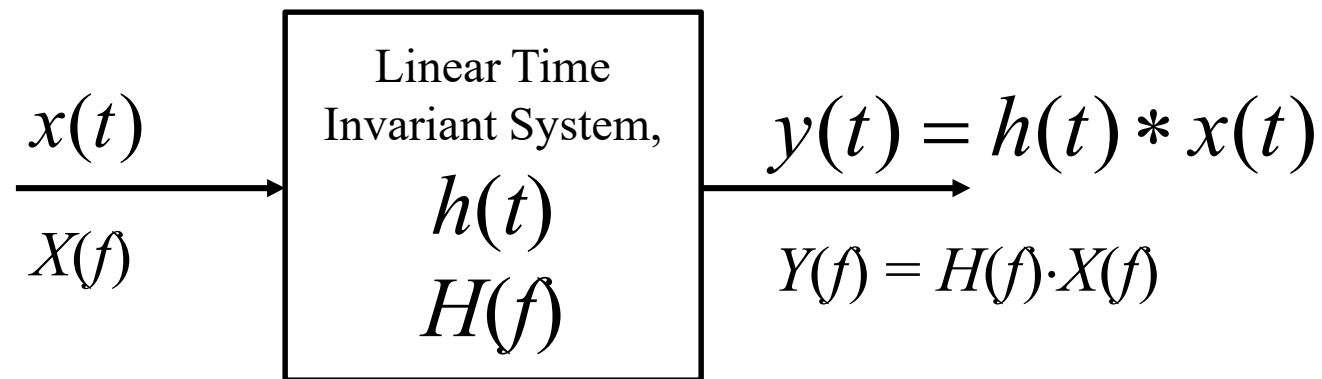
$$H(f) = |H(f)|e^{j\theta_h(f)}$$

$$\ln H(f) = \ln |H(f)| + j\theta_h(f) = \alpha(f) + j\theta_h(f)$$

- $\ln |H(f)|$ or $\alpha(f)$ is the **gain** of LTI system, measured in nepers

Signal Transmission through Linear Time Invariant System

Linear Time Invariant System



$$H(f) = |H(f)|e^{j\theta_h(f)}$$

$$\alpha'(f) = 20 \log_{10} |H(f)| \text{ dB}$$

- $\alpha'(f)$ is an alternate measurement of **gain** of LTI system, measured in dB

Signal Transmission through Linear Time Invariant System

Causal LTI System

- does not respond before the excitation is applied
- physically realizable system
- impulse response must vanish for negative time

$$h(t)=0, \quad t < 0$$

Signal Transmission through Linear Time Invariant System

Causal LTI System

Necessary and sufficient condition for causal system

$$\int_{-\infty}^{\infty} \left(\frac{|\alpha(f)|}{1 + f^2} \right) df < \infty \quad \text{where} \quad \alpha(f) = \ln|H(f)|$$

Signal Transmission through Linear Time Invariant System

Causal LTI System

Necessary and sufficient condition for causal system

$$\int_{-\infty}^{\infty} \left(\frac{|\alpha(f)|}{1 + f^2} \right) df < \infty \quad \text{where} \quad \alpha(f) = \ln|H(f)|$$

Paley-Wiener Criterion

Signal Transmission through Linear Time Invariant System

Causal LTI System

Necessary and sufficient condition for causal system

$$\int_{-\infty}^{\infty} \left(\frac{|\alpha(f)|}{1 + f^2} \right) df < \infty \quad \text{where} \quad \alpha(f) = \ln|H(f)|$$

Paley-Wiener Criterion

Significance

- If $\alpha(f)$ satisfy P-W criterion, $\alpha(f)$ can be combined with $\theta_h(f)$ to find $h(t)$ for a causal system
- Infinite attenuation ($\alpha(f) = -\infty$ or $\ln|H(f)| = 0$) for a **band of frequencies** DOES NOT satisfy P-W criterion
- However, $\alpha(f) = -\infty$ allowed for a **discrete set** of frequencies

Signal Transmission through Linear Time Invariant System

Stable LTI System

- output signal is bounded for all bounded input signals.
- The criteria is also known as *bounded input–bounded output (BIBO) stability criterion*

Signal Transmission through Linear Time Invariant System

Stable LTI System

For **bounded** input signal, $x(t)$

$$|x(t)| < M \quad \text{for all } t$$

Signal Transmission through Linear Time Invariant System

Stable LTI System

For **bounded** input signal, $x(t)$

$$|x(t)| < M \quad \text{for all } t$$

Output from a LTI system

$$y(t) = h(t) * y(t) = \int_{-\infty}^{\infty} h(t - \tau)x(\tau)d\tau = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)d\tau$$

Signal Transmission through Linear Time Invariant System

Stable LTI System

For bounded input signal, $x(t)$

$$|x(t)| < M \quad \text{for all } t$$

Output from a LTI system

$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(t - \tau)x(\tau)d\tau = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)d\tau$$
$$|y(t)| = \left| \int_{-\infty}^{\infty} x(t - \tau)h(\tau)d\tau \right| \leq \int_{-\infty}^{\infty} |x(t - \tau)||h(\tau)|d\tau$$

Signal Transmission through Linear Time Invariant System

Stable LTI System

For bounded input signal, $x(t)$

$$|x(t)| < M \quad \text{for all } t$$

Output from a LTI system

$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(t - \tau)x(\tau)d\tau = \int_{-\infty}^{\infty} x(t - \tau)h(\tau)d\tau$$

$$|y(t)| = \left| \int_{-\infty}^{\infty} x(t - \tau)h(\tau)d\tau \right| \leq \int_{-\infty}^{\infty} |x(t - \tau)||h(\tau)|d\tau \leq M \int_{-\infty}^{\infty} |h(\tau)|d\tau$$

Signal Transmission through Linear Time Invariant System

Stable LTI System

For bounded input signal, $x(t)$

$$|x(t)| < M \quad \text{for all } t$$

Output from a LTI system

$$|y(t)| \leq M \int_{-\infty}^{\infty} |h(\tau)| d\tau$$

Signal Transmission through Linear Time Invariant System

Stable LTI System

For bounded input signal, $x(t)$

$$|x(t)| < M \quad \text{for all } t$$

Output from a LTI system

$$|y(t)| \leq M \int_{-\infty}^{\infty} |h(\tau)| d\tau$$

output signal $y(t)$ to be bounded, $|h(t)|$ must be integrable, that is,

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$

Signal Transmission through Linear Time Invariant System

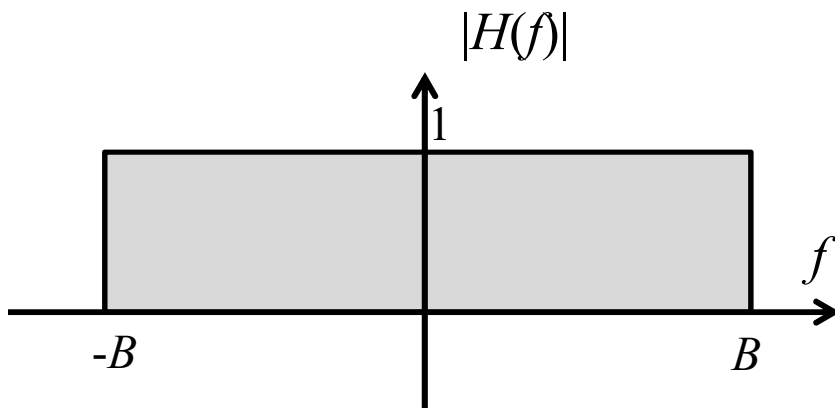
Stable LTI System

Necessary and sufficient condition for BIBO stability

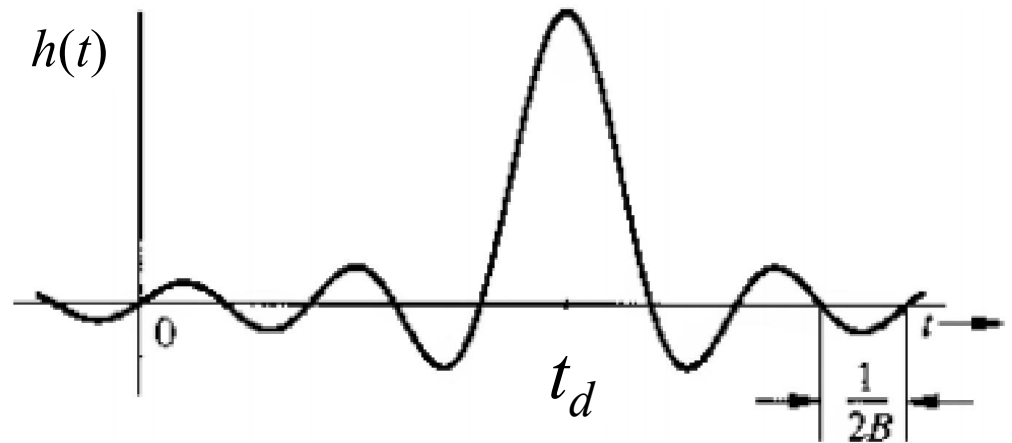
$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$

Signal Transmission through Linear Time Invariant System

System Response in
Frequency domain

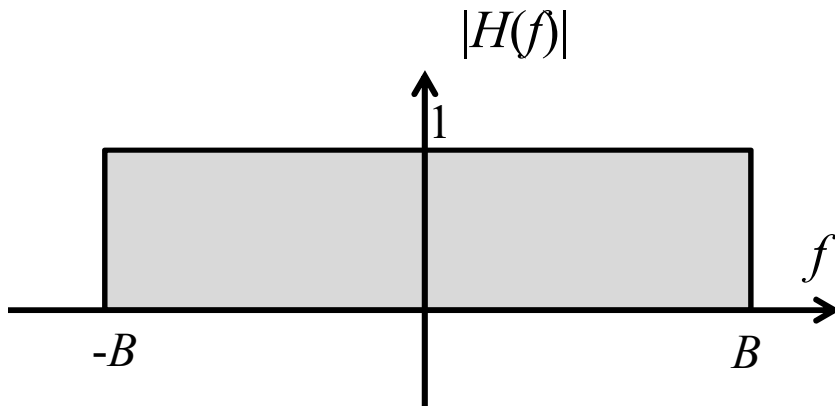


System Response in
Time domain

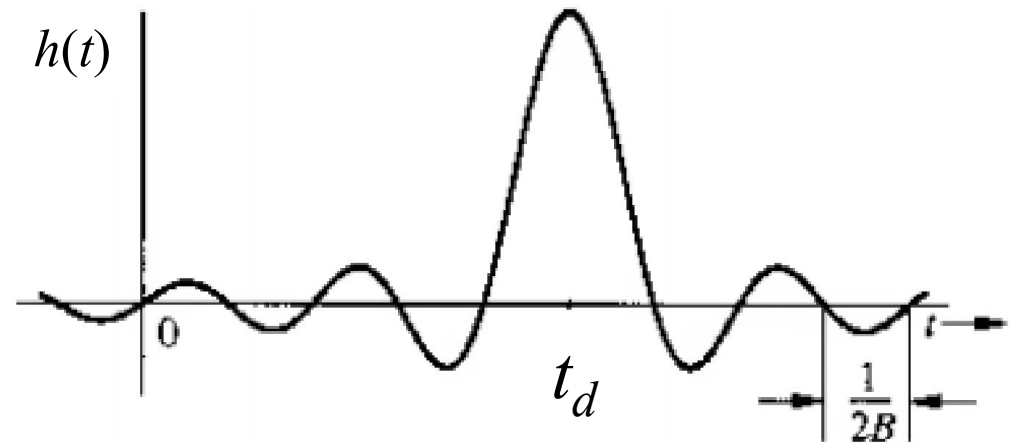


Signal Transmission through Linear Time Invariant System

System Response in
Frequency domain



System Response in
Time domain



Is this a causal system?